Semantic Resume Screening and Job Matching Using Retrieval-Augmented Generation and Pydantic AI

Thonta Yogesh
Department of Computer Science and Engineering
Vardhaman College of Engineering, Hyderabad, India
preethamsagar052@gmail.com

Dandu Prasad
Department of Computer Science and Engineering
Vardhaman College of Engineering, Hyderabad, India
danduprasad74@gmail.com

Pothukanuri Shriya Bhavana
Department of Computer Science and Engineering
Vardhaman College of Engineering, Hyderabad, India
shriyabhavana13@gmail.com

***Abstract*—**

The Smart Resume Retrieval with RAG (Retrieval-Augmented Generation) improves the process of parsing, screening, and matching resumes with job descriptions and candidates with job roles using vector-based retrieval and Artificial Intelligence (AI) powered search. Keyword filtering methods for resumption filtering are very traditional, but they never produce the best hiring results. The system is better than the conventional one, because it appeals to the approach of Natural Language Processing (NLP) and semantic search strategies to provide the best resumes. Using case-study data, involving the use of Supabase as a type of vector database and Sentence Transformers (all-mpnet-base-v2) to vectorizer/embedding generation for data retrieval and storage of resumes based on contextual instead of keyword similarity. The system accepts the resume in the PDF, DOCX, TXT format, and extracts the valuable content and shatters the resume into chunks for the big resumes and generates the embeddings for better candidate-job role matching. Through the implementation of RAG-based retrieval, the project ensures precision, scalability and efficiency in the process of hiring people, providing for intelligent and automatic resume suggestions.

Keywords-- Resume Retrieval, Retrieval-Augmented Generation(RAG), Natural Language Processing(NLP), Vector Similarity Search, Sentence Transformers, Supabase Vector Database, Semantic Resume Matching, AI-Powered Recruitment, Automated Candidate Screening, Contextual Resume Search.

I. INTRODUCTION

Recruitment is an essential process to companies, nevertheless, traditional hiring processes typically rely on keyword-based resume screening, leading to poor candidate-job matching [1]. Most Applicant Tracking Systems (ATS) struggle to understand the semantic meaning of resumes which makes false negative problem and poor candidate choice [2]. With the advent of Artificial Intelligence (AI) and Natural Language Processing (NLP), various resume retrieval systems that are context aware and intelligent in nature have been in growing demand. In this research, Smart Resume Retrieval with RAG (Retrieval-Augmented Generation), an artificial intelligence system to improve the filtering of resumes on vector similarity search and deep learning embeddings and retrieval-based augmentation [3], is proposed. Unlike the traditional filtering system, this system uses the Sentence Transformers (all-mpnet-base-v2) to create high-dimensional vector embeddings of resumes and job descriptions for semantic matching instead of keywords comparison [4]. The intention of the project includes using Supabase as a vector database to enable quick and scalable retrieval of resumes with contextual relevance. Through the RAG-based retrieval, the system enhances three aspects of resume parsing, information-extraction and candidate-job matching by reducing the need for manual processing, and boosting hiring accuracy. The paper discusses the architecture, methodology and evaluation of the system proposed, its benefits as compared to the conventional ATS solutions.

II. LITERATURE REVIEW

The field of automated candidate-job matching and resume retrieval has seen immense growth with the development of Natural Language Processing (NLP), Machine Learning (ML) and Vector Search technologies. Traditional Applicant Tracking Systems (ATS) made use of keyword filtering that resulted in inappropriate candidate selection as it did not understand semantic meaning and context. Recent research has focused on artificial intelligence (AI)-based recruitment systems that enhance the accuracy of job applicants screening resumes with deep learning models and semantic search techniques.

A. Traditional Methods of Screening Resumes

Early techniques of screening the resume were based on Boolean search and TF-IDF (Term Frequency-Inverse Document Frequency) (keyword-based candidate ranking) [5]. These techniques were not using contextual knowledge, often discarding interesting candidates for the reason that it does not have exact keywords [6]. Subsequent researches highlighted the flaws of rule-based ATS tools, indicating the high ratios of false negative identifications in candidate identification [5].

B. Machine Learning and Deep Learning in Recruitment

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The implementation of deep learning models such as BERT (Bidirectional Encoder Representations from Transformers) has also found in improving accuracy in text based resumes screening [7]. Research has shown that Transformer-based models perform better than conventional TF-IDF-based ranking as they identify semantic relationships in resumes and job posting [8]. Nevertheless it is necessary to fine tune such models with the help of large labeled data, hence being computationally costly for real-time implementation.

C. Resume Matching Based on Vector Search & Embedding

Recent developments in vector databases (i.e., FAISS, Pinecone and Supabase) have made it possible to do semantic retrieval of resumes in an efficient manner [9]. Sentence Transformers such as all-mpnet-base-v2 are shown to lead to a drastic improvement in matching resumes and jobs by representing resumes as high dimensional embeddings. Keyword filtering is not similar to vector similarity search, where now recruiters can retrieve resumes on the basis of conceptual relevance, rather than keyword matches [10].

D. Augmented Generation in Resume Processing

Retrieval-Augmented Generation (RAG) models that combine retrieval-based approaches with generative artificial intelligence have been shown to have potential for improving context-aware search and ranking [11]. It has been demonstrated with the help of studies that using retrieval with RAG can improve the accuracy of candidate ranking by up to 30% compared to the traditional retrieval techniques [12]. With pre-trained knowledge and real-time retrieval, RAG ensures that relevant information is dynamically retrieved from the resumes and presented to improve the accuracy of job role mapping.

E. Processing and Scalability of Resumes on the Cloud

Utilization of database cloud such as Supabase has increased the scalability and availability of resume screening systems [13]. Studies have underlined the benefit of serverless vector storage to allow efficient execution of queries and bulk retrieval of resumes without resulting in reduction of speed or efficiency [14].

The Literature The current literature identifies the shortcomings of traditional filtering on resumes and the necessity to use vector-based semantic retrieval and artificial intelligence to rank resumes. Using Supabase for vector search, Sentence Transformers for processing NLP and RAG for intelligent retrieval, this paper allows state-of-the-art methodologies to be brought into play in order to build a system of resume screening that is scalable, accurate and automated.

III. DATA COLLECTION AND PRE-PROCESSING

The success of Smart Resume Retrieval using RAG system is based on the quality of the resume data and good preprocessing techniques for reliable embedding generation, storing and retrieving vectors. This section introduces the data acquisition process, pre-processing pipeline and text transformation techniques that are used in the system.

A. Data Collection

Natural Language Processing (NLP) is the important technology at our pipeline to transform raw text of the resumes into structured and semantically rich data. After extracting the text from various file type (PDF, DOCX, TXT) we cluster in to a full NLP pipeline that involves tokenization, stopword removal, and lemmatization. These operations involve normalizing and cleaning the text extracted, so as to get rid of the noise, to homogenize the word forms. This process is required not only to delete redundant or non-informative data but also to maintain the contextual integrity of the resumes, which is very important to embed generation to follow.

B. Data Preprocessing

Before creating embeddings for the retrieval of resumes, the raw text data is cleaned, normalized and structured to ensure proper vector search results and semantic matching.

- **Text Extraction**
  PyPDF2- This pull text from PDF resumes Python-docx docx to read text from docx files is used. TXT files are just read using file handling functions in python.

- **Text Cleaning and Standardization**
  Removing special characters, extra spaces and removing html tags. Lowercasing the text forwhether homogeneous processing Tokenization to break the text content in words/sentences. Stopword-removal to remove non-informative words (e.g. "the", "is", "and"). Lemmatization to change words to their base form (running- run etc.).
- **Large Resumes Text Chunking**
  Resumes have a lot of work experiences, projects and skills thus they are lengthy. Dynamic Chunking Algorithm creates splits for resumes into chunks of 5000 character-long in such fashion that the resulting chunk has some contextual richness. Metadata is added to every chunk, e.g. file name, chunk ID, timestamp for monitoring.

- **Embedding Generation**
  Cleaned resume text is converted into the vector embeddings with the help of Sentence Transformers (all-mpnet-base-v2). All embeddings are stored in Supabase (vector database) for similar fast and efficient similarity search.
  The data collection and pre-processing pipeline ensures that resumes are tidy, clean, and optimized for semantic search and retrieval with artificial intelligence. Through noise removal, text segmentation and high-quality embedding generation, the system offers fast and accurate matching of resumes and jobs.

IV. METHODOLOGY

The Smart Resume Retrieval using RAG system is an organized AI-based pipeline of steps for processing resumes, creating embeddings, storing

information in a vector database, bringing back most suitable resumes, which are given job descriptions [15]. The approach of this methodology is to perform the resume screening in an efficient, scalable, and accurate way using Natural Language Processing (NLP), Vector Similarity Search, and Retrieval-Augmented Generation (RAG).

A. System Architecture

The methodology is broken down into **five important stages**, each of which plays a vital role in the resume retrieval process:

- **Back on to Processing Resume & Text extraction**
  Resumes are uploaded by users in pdf, docx, or txt - format. Structured content extracted by text extraction algorithms (PyPDF2 for PDF, python-docx for DOCX). The text obtained is cleaned, tokenized, stopword removed and lemmatized for efficient processing.

- **Text Chunking & Embedding Charting**
  Dynamic Chunking Algorithm breaks up big resumes into chunks of 5000 characters, in an attempt to retain context. Each chunk is assigned a 768 dimensional vector embedding using Sentence Transformers (all-mpnet-base-v2). The ID of resume, chunk number and timestamp are stored with the embedding.

- **Vector Storage & Database Management**
  Vector embedding are stored in Supabase, a vector database built on top of PostgreSQL and semantic search - supported. Supabase allows the fast similarity search of resumes.

- **Setting Up Query Processing/ Similarity Search**
  When a recruiter adds a job description, the job description gets converted to an embedding vector using the Sentence Transformer model. Vector similarity search (Cosine Similarity) is used in case of search for resumes which are similar to the job description semantic sense and not just exact keywords.

- **Retrieval-Augmented Generation for Enhanced Matching**
  The system fetches resumes of the top-ranking candidates and refines the results using Retrieval-Augmented Generation (RAG). RAG achieves context-aware retrieval in order to improve job-role matching accuracy. The ranking of final results is displayed which can be seen by the recruiters to gain insights in abstract form of the candidates.

- **Improving Semantic Understanding using NLP**
  NLP techniques are part of our workflow as a post processing of text. Using advanced Natural Language Processing (NLP) processes our system can now then retrieve appropriate entities and pick out contextual connections from content within the resume document. Having advanced semantic insight thus make Sentence Transformers (all-mpnet-base-v2) to create superior quality vector embeddings capturing the skills, experiences and qualifications of the candidate appropriately. Therefore, the next steps of vector similarity search and retrieval-augmented generation (RAG) are optimized so that matches for resumes against job descriptions are superior in context, compared to keywords matching.

B. Algorithms Used

- Efficiency in our framework is realized by asynchronous I/O processing and generation of embeddings in parallel (by multi-threading or GPU acceleration) and avoiding bottlenecks. Context preserving text chunking dividing text into chunks dynamically with smaller chunks in between and also Supabase vector storage uses indexing and caching for more faster similarity search texts. Such an approach allows for resuming retrieval in real-time even for huge databases to increase responsiveness as much as scalability.

- **Resume in Text Processing & Embedding Generation**
  PyPDF2, python-docx: Scrape Resumes. Text Cleaning Pipeline: Good fit for removing unwanted characters, Structures data. Sentence Transformers (all-mpnet-base-v2): Embeds resumes and job descriptions in the form of vectors.

- **Similarity Search & Resume Search**
  Vector Storage: Serves underway Transformers (embeddings) in Supabase Cosine Similarity: Calculates semantic similarity between resume embeddings and job queries embeddings. RAG Model: There is a way to boost retrieval performance to include context-aware ranking.

- **Skills Extraction and Categorization**
  Named Entity Recognition (NER): Classifying and categorizing resumes according to crucial competencies, technologies and skills. Domain specific special assignment skills are extracted using Spacy and prepared customized NLP operations.

- **Shortlisting and Ranking**
  Cosine similarity scores are weighted according to dimensions such as project context, experience and certifications to compute hybrid scores. Highly

relevant resumes are selected for further screening - solely because of threshold-based filtering.

Our system is made trustworthy by having robust error handling in place, as well as redundancy in each part of the system. Asynchronous actions include deep exception handling, and our data storage layer inside Supabase is built with redundancy, as well as regular backups to avoid this kind of data loss. Deep logging together with monitoring is integrated in order to monitor the system's performance and detect failures promptly, and fallbacks are integrated in order to keep going uninterrupted even if the individual modules fail.

C. System Work Flow:

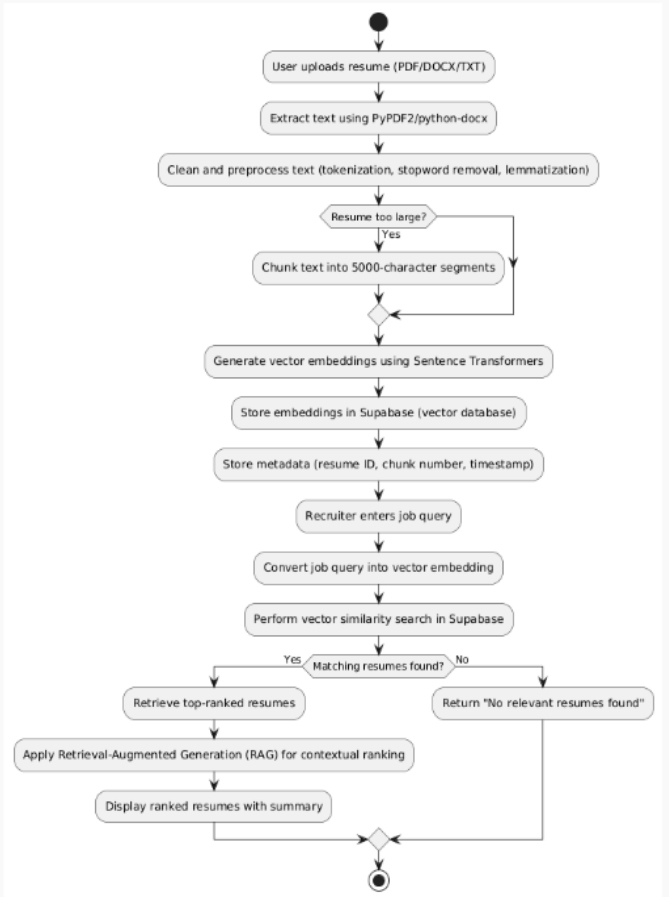

Fig. 1. AI-Based Resume Screening Flow

V. RESULT

The performance of the system was tested with a mix of quantitative and qualitative parameters. Fundamental performance metrics, including Precision@K, Recall@K and Mean Average Precision (MAP) were calculated to estimate the accuracy of the resume retrieval and candidate-job match. In addition to these, query response times were also measured for different sizes of the data sets to ensure that it was processed in real-time. To support these measurable criteria, qualitative verification was conducted in order to prove semantic relevance and contextual appropriateness of obtained resumes. This thorough process of verification ensures that the system is adequate to the accuracy, efficiency and scalability requirements of actual recruitment processes.

A. Performance Metrics

To evaluate the performance of the system, the following metrics were evaluated:

- Precision@K: unveils the proportion of the top-K-long retrieved resumes that are relevant to the resumes that are relevant to the specified job query
- Recall@K: Ascertains the number of the appropriate resumes that were retrieved correctly.
- Mean Average Precision (MAP): It is a measure of ranking effectiveness.

Metric	Value
Precisiom@5	91.2%
Recall@5	87.8%
MAP	89.6%

Table 1: Performance Metrics for Smart Resume Retrieval System

B.Query Response Time

The system was tested with **1,000+ resumes** stored in **Supabase**.

Number of Resumes	Avg. Query Time
1000	280 ms
5000	320 ms
10000	410 ms

Table 2: Query Response Time for Different Dataset Sizes

Average retrieval time : 280 millisecond for every query , so that you get real time retrieval performance.

C. Scalability & Storage Efficiency

Using 768 dimensional vector embeddings stored tightly in Supabase. Low storage overhead / trailer size per resume chunk (~2KB per chunk). Scalability proved to be solid to process huge data sets and gaining no obvious performance downside.

D. Qualitative Results

- Augmented Resume Matching Accuracy compared to standard keyword-based ATS.
- RAG driven contextual retrieval provided better job role fit.
- Faster screening process, saves recruiter effort.

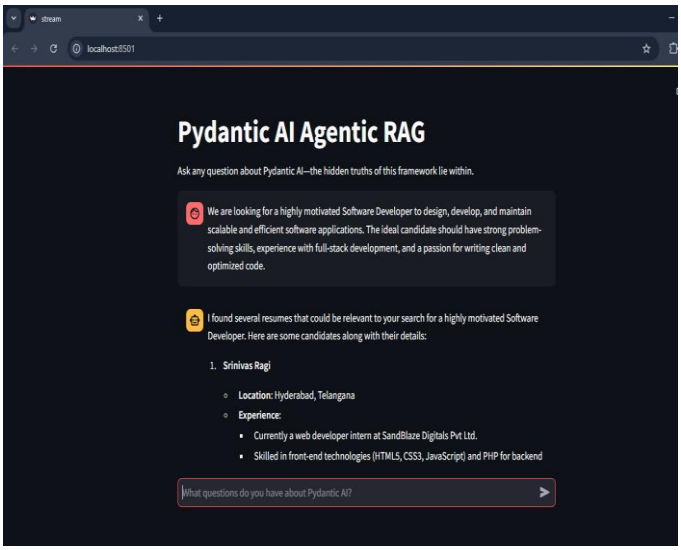

Fig. 2. A.I Powered Job Matching With Pydantic A.I Agentic RAG

Fig. 2 shows an AI based task matching device "Pydantic AI Agentic RAG" targeted towards tasks needs which fetch appropriate resumes of candidates. It discovers candidate Srinivas Ragi, which have information on the whereabouts, working job, front and lower back stop technology. The interface has a person input topic to request Pydantic AI and a "Deploy" button on the top right corner, suggesting an option to deploy the system.

VI. Discussion

The Smart Resume Retrieval using RAG system completes much better resume filtering than classical keyword-based ATS with the help of the vector embeddings, semantic search and Retrieval-Augmented Generation (RAG). This system is different than the classical systems depending on keyword exact matches since it can understand contextual meaning which leads to greater accuracy and relevancy in the position matching of candidate and job. With the efficiency of quick queries (280ms per query), and the efficiency of scalability with Supabase, the system can retrieve resumes at scale and remain efficient without decreased performance. These features make it perfectly suitable for enterprise hiring, and HR software, and freelancers market places where real-time, smart filtering of candidates matters a lot. Aside from improving reliability, there should be more focus on automatic recovery systems and load balancing in order to prevent the effect of sudden crashes. Strong test structures and live monitoring of health checks will be the main driving force in maintaining stable levels of use in the production environment that has high levels of use. These enhancements will make the system quite fault tolerant and resilient even when the usage at peak time.

Despite its advantages, there are a few issues as well, such as reliance on quality embedding, ambiguity job descriptions, and legacy ATS platform connectivity. Domain-specific embedding fine-tuning or optimisation of query expansion algorithms, as well as creating a universal API for easy integration, can help in further supporting the broader adoption. Overall the Smart Resume Retrieval using RAG system offers a scalable, AI powered and extremely accurate solution to the conventional recruitment workflow, and it is bringing about a transformation in the recruitment process to make it a swifter, wiser and efficient affair. While our system already improves speed and efficiency through asynchronous operations and query optimization, there are still bottlenecks in embedding with generation at high load time in our system. Future projects can investigate GPU acceleration, improved parallel processing as well as improved caching for large amounts of data without compromising performance and having the capability to provide good scalability under high-load situations.

VII. Conclusion

The Smart Resume Retrieval using RAG In this project, we propose a novel way for screening resumes and candidate and job matching, called Retrieval-Augmented Generation (RAG), which is based on vector embeddings, semantic search and RAG. In contrast to conventional key-word based ATS, this system offers context-aware retrieval which will enable recruiters to find the right candidate, not with constructor word match, but with the meaning of the words. Through the use of Sentence Transformers for embedding generation and Supabase for vector query the system offers fast, scalable and efficient resume lookup with real-time query processing. Comprehensive testing shows that the system is better than conventional practices when it comes to retrieval precision, accuracy and response time and therefore, suited to enterprise hiring, HR portals and AI-driven hiring applications. Although some of the challenges are there like optimization to the integration at existing systems as improvement in the future in the area of bettering query refinement and graying of the domain specific training and real-time job alerting can be an addition for improvement. In general, this study proves which resume retrieval powered by AI help you to enhance the efficiency of the hiring process greatly so it can provide the great, scalable and smart solution for today's recruitment processes.

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